MD ASHIKUL HAQUE

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Google Scholar: scholar.google.com/citations?user=11vTHPYAAAAJ

EDUCATION

PhD, Computer Science Expected May 2027
University of Texas at Dallas, USA
Thesis: *Resilient and Secure LPWAN Communication Under Jamming and Coexistence*
Advisor: Prof. Abusayeed Saifullah

MSc, Computer Science 2024
Wayne State University, USA

BSc, Computer Science and Engineering 2018
Bangladesh University of Engineering and Technology, Bangladesh

RESEARCH INTERESTS

LPWAN and Wireless IoT Systems • Resilient and Secure Communication • MAC and Scheduling • AI-Assisted Coexistence Management

Research Summary. I develop systems, protocols, and AI-assisted methods for LPWAN and wireless IoT networks. My work focuses on resilient and adaptive communication under interference, coexistence, and jamming, including LoRa anti-jamming defenses, learning-based coexistence management, cross-technology communication from LR-FHSS to LoRa, and MAC and scheduling mechanisms for bursty, reliable, and time-sensitive traffic in IoT and cyber-physical settings. I validate these designs through a combination of PyTorch-based learning, NS-3 simulation, SDR prototyping with GNU Radio and USRP, and outdoor or testbed experiments.

CONFERENCE PUBLICATIONS

[9] Md Ashikul Haque, Abusayeed Saifullah. *Deadline-Aware Token Scheduling for Physical-I/O LLM Agents*. In Proc. IEEE Real-Time Systems Symposium (RTSS 2026), pp. 1–12. Acceptance rate: 14%.

[8] Md Ashikul Haque, Aakriti Jain, Venkata Prashant Modekurthy, Abusayeed Saifullah. *Enabling Cross Technology Communication from LR-FHSS to LoRa*. In Proc. ACM/IEEE International Conference on Embedded Networked Sensor Systems (SenSys 2026), CPS-IoT Week 2026. Acceptance rate: 19.8%.

[7] Md Ashikul Haque, Abusayeed Saifullah. *Decoding LoRa Packets under Collaborative Jamming Attacks*. In Proc. 23rd International Conference on Embedded Wireless Systems and Networks (EWSN 2026). Acceptance rate: 23.5%.

[6] Md Ashikul Haque, Abusayeed Saifullah. *ReMix: Resource-Reclaiming Mixed-Criticality Scheduling for Industrial IoT*. In Proc. The 35th International Conference on Computer Communications and Networks (ICCCN 2026), pp. 1–10. Acceptance rate: 25%.

[5] Md Ashikul Haque, Abusayeed Saifullah. *Mitigating Jamming Attacks in LoRa Networks: A Defense Strategy Against LoRa Based Jammers*. In Proc. ACM International Symposium on Mobile Ad Hoc Networking and Computing (MobiHoc 2025). Acceptance rate: 23%.

[4] Md Ashikul Haque, Abusayeed Saifullah, Haibo Zhang. *Deep Reinforcement Learning Based Coexistence Management in LPWAN*. In Proc. IEEE Conference on Computer Communications (INFOCOM 2025). Acceptance rate: 18.6%.

[3] Md Ashikul Haque, Abusayeed Saifullah. *Handling Jamming Attacks in LoRa Networks*. In Proc. ACM/IEEE Conference on Internet of Things Design and Implementation (IoTDI 2024), CPS-IoT Week 2024.

[2] Aakriti Jain, Md Ashikul Haque, Abusayeed Saifullah, Haibo Zhang. *Burst-MAC: A MAC Protocol for Handling Burst Traffic in LoRa Network*. In Proc. IEEE Real-Time Systems Symposium (RTSS 2024).

[1] Md Ashikul Haque, Abusayeed Saifullah. *A Game-Theoretic Approach for Mitigating Jamming Attacks in LPWAN*. In Proc. International Conference on Embedded Wireless Systems and Networks (EWSN 2023).

WORKSHOP PUBLICATIONS

[1] Md Ashikul Haque, Abusayeed Saifullah, Haibo Zhang. *Coordinating Coexisting Learning Agents in Shared Spectrum via Parameter Space Complementarity*. Agents in the Wild: Safety, Security, and Beyond (ICLR 2026).

[2] Md Ashikul Haque, Dali Ismail, Abusayeed Saifullah. *Scalable Real-Time Control in Industrial Cyber-Physical Systems*. In Proc. Workshop on Software-Defined Networking for Cyber-Physical System Automation (SDN-CPS 2024),

co-located with ICDCN 2024, pp. 353–358.

Manuscripts Under Review

Md Ashikul Haque, Abusayeed Saifullah, Haibo Zhang.
Generative Policy Coordination for Coexisting Learning-Enabled LPWANs.

Selected Talks

Mitigating Jamming Attacks in LoRa Networks: A Defense Strategy against LoRa Based Jammers Oct 2025
ACM MobiHoc 2025, Rice University, Houston, USA

Handling Jamming Attacks in a LoRa Network May 2024
ACM/IEEE IoTDI 2024, Hong Kong, China

Research and Professional Experience

Graduate Research Assistant, Dept. of Computer Science 2022 to Present
The University of Texas at Dallas, USA | Advisor: Prof. Abusayeed Saifullah

- Designed and implemented cross-technology communication mechanisms from LR-FHSS to LoRa using GNU Radio and USRP, enabling long-range downlink and control for LPWAN deployments.
- Developed signal-processing and protocol mechanisms to improve LPWAN resilience against reactive and collaborative jamming, with evaluation on real testbeds and outdoor experiments.
- Built learning-based coexistence management and telemetry-driven evaluation pipelines for LPWANs using PyTorch, NS-3, and measurement-driven experimentation.

Assistant Engineer, ICT Division 2020 to 2021
Dhaka Electric Supply Company, Bangladesh

- Maintained ICT infrastructure with monitoring, incident response, and reliability focused operations for distributed utility systems.

Software Engineer, IoT Division 2019 to 2020
Samsung R&D Institute Bangladesh, Bangladesh

- Developed production grade C++ components for Samsung Cloud client integration supporting SmartThings workflows and IoT device communication.
- Delivered iOS integration via Swift wrapper and onboarding features, collaborating across client libraries and application layers.
- Automated multi step build and packaging workflows using Bash to improve reproducibility and reduce manual release effort.

Teaching

Graduate Teaching Assistant, Dept. of Computer Science 2026
The University of Texas at Dallas, USA
Courses: Computer Networks; Performance of Computer Systems and Networks

- Graded assignments, exams, and project deliverables for Computer Networks and Performance of Computer Systems and Networks.
- Helped students understand course concepts through office hours, question answering, and discussion of networking and systems topics.
- Supported course projects by clarifying requirements, troubleshooting implementations, and providing feedback on design and debugging.

Guest Lecturer (multiple sessions) 2025
The University of Texas at Dallas, USA
Course: Internet of Things

Lecturer, Dept. of Computer Science 2018 to 2019
Millennium University, Bangladesh
Courses: Structured Programming; Data Communication

Awards and Grants

Dean's List Award, Wayne State University 2024
Professional Travel Award, College of Engineering, Wayne State University 2024
SIGBED Travel Grant, CPS-IoT Week, San Antonio, Texas, USA 2023

Professional Service

Conference Referee

- ACM/IEEE Intl. Conf. on Embedded Networked Sensor Systems (SenSys) 2022, 2024, 2025, 2026
- IEEE Real-Time Systems Symposium (RTSS) 2023, 2024, 2025
- IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS) 2023, 2024
- Intl. Conf. on Embedded Wireless Systems and Networks (EWSN) 2023, 2024, 2025
- IEEE Intl. Conf. on Distributed Computing in Smart Systems and IoT (DCOSS) 2024

Conference Reviewer

- IEEE Real-Time Computing Systems and Applications (RTCSA) 2024, 2025
- IEEE Intl. Conf. on Sensing, Communication, and Networking (SECON) 2026
- IEEE Intl. Conf. on Distributed Computing in Smart Systems and IoT (DCOSS) 2026

Journal Referee

- IEEE/ACM Transactions on Networking 2024, 2025

Journal Reviewer

- Pervasive and Mobile Computing (Elsevier) 2023

Volunteer: CPS-IoT Week 2023

Technical Skills

Programming: C, C++, Python, MATLAB, SQL, Swift, Bash
ML and Data: PyTorch, Pandas
Wireless: GNU Radio, SDR pipelines, Telemetry
Simulation: NS-3, Linux, Docker, Git, LaTeX
Hardware: USRP B200, LR-FHSS, LoRa and LPWAN devices, IoT gateways

References

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